

1.Program Using %TYPE

DECLARE

a type.TA %TYPE;

b type.DA %TYPE;

t type.total %TYPE;

BEGIN

SELECT TA, DA INTO a, b FROM type
WHERE emp_id = 20;

t := a + b;

UPDATE type SET total = t WHERE emp_id
= 20;

DBMS_OUTPUT.PUT_LINE('Total
Updated: ' || t);

END;

/

2.Program Using %ROWTYPE

DECLARE

emp_record emp%ROWTYPE;

```
BEGIN
```

```
    SELECT * INTO emp_record FROM emp  
WHERE id = 1;
```

```
    DBMS_OUTPUT.PUT_LINE('Employee ID: '  
|| emp_record.id);
```

```
    DBMS_OUTPUT.PUT_LINE('Name: ' ||  
emp_record.name);
```

```
    DBMS_OUTPUT.PUT_LINE('Salary: ' ||  
emp_record.salary);
```

```
END;
```

```
/
```

3.Explicit Cursor Example

```
DECLARE
```

```
    CURSOR emp_cur IS SELECT id, name,  
salary FROM emp;
```

```
    v_id emp.id%TYPE;
```

```
    v_name emp.name%TYPE;
```

```
    v_salary emp.salary%TYPE;
```

```
BEGIN
```

```
OPEN emp_cur;  
LOOP  
    FETCH emp_cur INTO v_id, v_name,  
v_salary;  
    EXIT WHEN emp_cur%NOTFOUND;  
    DBMS_OUTPUT.PUT_LINE('ID: ' || v_id || ',  
Name: ' || v_name || ', Salary: ' || v_salary);  
END LOOP;  
CLOSE emp_cur;  
END;  
/
```

4.User-Defined Exception Example

```
DECLARE  
    insufficient_balance EXCEPTION;  
    v_balance NUMBER := 500;  
    v_withdraw NUMBER := 1000;  
BEGIN  
    IF v_withdraw > v_balance THEN  
        RAISE insufficient_balance;
```

ELSE

 v_balance := v_balance - v_withdraw;

END IF;

 DBMS_OUTPUT.PUT_LINE('Remaining
Balance: ' || v_balance);

EXCEPTION

 WHEN insufficient_balance THEN

 DBMS_OUTPUT.PUT_LINE('Error:
Withdrawal exceeds balance.');

END;

/

5.Procedure Example

CREATE OR REPLACE PROCEDURE

calc_square (p_num IN NUMBER, p_result
OUT NUMBER) AS

BEGIN

 p_result := p_num * p_num;

```
END;
```

```
/
```

```
DECLARE
```

```
    v_result NUMBER;
```

```
BEGIN
```

```
    calc_square(6, v_result);
```

```
    DBMS_OUTPUT.PUT_LINE('Square is: ' ||  
v_result);
```

```
END;
```

```
/
```

6.Trigger Example

```
CREATE TABLE main (id NUMBER, salary  
NUMBER);
```

```
CREATE TABLE backup (id NUMBER,  
salary NUMBER);
```

```
CREATE OR REPLACE TRIGGER
```

```
trg_backup_main
```

```
BEFORE DELETE ON main
FOR EACH ROW
BEGIN
    INSERT INTO backup VALUES(:OLD.id,
:OLD.salary);
END;
/
```

7.VARRAY Example

```
DECLARE
    TYPE NumberArray IS VARRAY(5) OF
NUMBER;
    nums NumberArray := NumberArray(10,
20, 30);
BEGIN
    FOR i IN 1..nums.COUNT LOOP
        DBMS_OUTPUT.PUT_LINE('Element ' || i
|| ':' || nums(i));
    END LOOP;
END;
```

